

Campus Wayfinder

Ask out loud, “How do I get to PSB 120?”, and get indoor directions.

The Idea

Google Maps gets you to the building, then you’re lost inside. Campus Wayfinder handles the indoor part: rooms, stairs, elevators, and entrances. You ask by voice, it plans a route through a map of the building, and Grok Voice walks you through it using landmarks (*“Past the atrium, take the stairs on your left”*). Grok Imagine can generate a simple illustrated step card for each turn. The demo can happen in the hackathon building itself.

Track Fit

Built with Cursor	Map editor, routing, and UI built in Cursor.
Grok Voice / Imagine	Voice directions and questions; optional Imagine step cards.
Space data	Not used.
Navigation theme	Direct fit: indoor navigation.
Also eligible	Big Red, Software, Design, People’s Choice (every hacker has been lost in PSB).

Features

MVP

- Hand-built map of PSB (and Klarman) as a graph: rooms, hallways, stairs, elevators.
- Voice query to route: *“Take me to the career fair.”*
- Shortest path on the graph, spoken step by step with landmarks.
- “Step-free” option that uses elevators only.

Stretch

- Event-aware: *“Where’s the next workshop?”* using the schedule.
- QR codes at hallway corners to confirm where you are.
- Illustrated step cards from Grok Imagine.

Tech Stack and Data

- Map: a JSON graph drawn over the PSB floor plan (hand-made).
- Routing: Dijkstra/A* on the graph (small, runs in the browser).
- Frontend: React; SVG floor plan with the route highlighted.
- AI: Grok Voice; Grok Imagine (optional).

Limitations and Risks

- **No indoor map data.** There's no public, routable indoor map of PSB. Someone must walk the building and draw it.
Mitigation: Map only the floors judges will see. Split the work: one person maps while others code.
- **No indoor positioning.** GPS doesn't work well inside buildings, so the app can't know where you are automatically.
Mitigation: Ask the user ("I'm at the atrium") or use QR codes at key spots.
- **Does not scale.** Every new building needs a hand-drawn map.
Mitigation: Pitch it as a platform: show the simple map editor and how fast a new building can be added.
- **Generated images can mislead.** Grok Imagine pictures won't look like the real hallway.
Mitigation: Keep them as stylized icons, or use real photos instead.
- **Less technical depth.** Graph routing is simple; Software judges may want more.
Mitigation: Add depth with step-free routing, schedule awareness, and natural-language queries.

Scorecard

Demo-able	5/5	The demo is in the building you mapped.
Buildable in 24h	4/5	Map drawing is tedious but simple.
Navigation theme	5/5	Indoor navigation.
Wow factor	3/5	Practical more than flashy.
Technical risk (5 = riskiest)	2/5	Low.

Verdict

Safe, crowd-pleasing pick. Strong for People's Choice and the live demo; needs a creative twist to win on technical depth.